

ARTISTS HOUSING ACHIEVES ENERGY STAR PERFORMANCE

Artists converting old industrial buildings into studio live/work spaces is a familiar story in many urban areas, but the conversion of a decrepit former factory into Brookside Artists Housing came with numerous twists and turns.

BY KIMBERLY VERMEER

At the groundbreaking ceremony for Brookside Artists Housing in Boston, a city councilor recalled how she had spent hours on an Election Day three years earlier sitting with the artists at a hearing that prevented the building's owner from evicting them. This was a key victory, but it was only the first of many legal and political steps that were needed over several years in order for the artists to secure ownership of the building. Once they owned the building, the artists could address the significant rehabilitation that was needed to bring it up to code, and to transform it into safe and secure live/work spaces. The artists succeeded in getting the building not only up to code—but up to Energy Star standards. Budgetary problems nearly quashed their Energy Star-related aspirations; a creative use of funding sources was what pulled it off.

The artists brought in New Atlantic Development Corporation (NADC)—an experienced developer with several urban and historic rehabilitation projects in its portfolio—to oversee the project (see “Achieving High Performance in Affordable Multifamily Housing,” *HE* May/June '04, p. 34).

When completed, Brookside will have 24 units. Three units are commercial/work-only spaces, but the remaining 21 are live/work spaces under a special provision of the Boston zoning code that allows combined commercial and residential uses for



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artist housing. Three of the units will be sold at market rates; three will be sold at prices affordable to artists earning 60% or less of the median income for the area; and the rest will be sold at prices affordable to artists earning 80% or less of the median income. The development of Brookside is part of the city of Boston's Affordable Artist Housing program. In order to purchase a unit, applicants must demonstrate that they are working artists (deriving at least 10% of their income from art) whose income does not exceed the program limits. The units carry deed restrictions that preserve their use as artist live/work spaces and maintain their affordability over time.

Energy and Health Concerns

The buildings—post-and-beam construction with wide board sheathing, randomly spaced wall studs, and stone foundations—needed a total gut rehab for conversion into safe and comfortable live/work spaces. Peter Roth, president of NADC, was familiar with the Energy Star Homes program from previous projects, and he wanted to develop an Energy Star-rated project to provide the artists and other residents with energy-efficient homes and studios. The plans used technical standards from the Energy Star Homes program and called for insulation, air sealing, new windows, and energy-effi-

cient heating and hot water systems. However, it was difficult to determine the best heating system for the small (800–1,600 ft²) units, since most of the available options were oversized.

Options were further complicated by the condominium ownership structure that would be the end result of the renovation and by the artists' desire to allocate costs to individual units while keeping common expenses and condominium fees to a minimum. Ultimately, we chose three Rinnai sealed combustion, gas-fired, variable output heating units. With capacities of 9,000 Btu/hr (81% annual fuel utilization efficiency or AFUE), 18,000 Btu/hr (84% AFUE), and 31,500 Btu/hr (82% AFUE), the Rinnai units could be mixed and matched for the unit size and provided the best option for cost effectiveness and relatively high efficiency.

In keeping with its industrial past and the size of the openings in the building, replacement windows were custom made. The metal-framed double-glazed windows, with low-e coatings and argon fill, have a U-factor of 0.54. The plans also called for Energy Star-rated refrigerators, dishwashers, and CFL interior and exterior lighting.

The development's address on Brookside Avenue provides a clue to one of the major health concerns: moisture management. The original construction included stone foundations and unfinished crawlspaces that were often wet, due to the nearby brook, the resulting high water table, and poor drainage. The health implications of creating airtight live/work spaces over wet, smelly crawlspaces were a worry, especially once constant exhaust fans were installed in the units to meet Energy Star Homes requirements. The rehabilitation plan called for new drainage to reduce future moisture problems, but it did not call for concrete floors. What kind of old industrial dust and bacterial contaminants would get pulled up through the old floorboards once the rest of the envelope was air sealed and the exhaust fans were turned on?

Resource Issues

Projects always have budget issues, and a project for "starving artists" is no exception. The renovation budget had to address basic code requirements as a first priority. The plan called for insulation, energy-efficient equipment, and air sealing to meet Energy Star Homes standards, but these standards are designed for new construction. As construction got underway, the Energy Star Homes program staff were worried that the existing conditions were so difficult that it would be impossible to install fiberglass batts properly or to meet the air sealing standards of 2 square inches or less leakage per 100 square feet of surface area to achieve the base qualification score of 86 using REM:Rate software.

The project was already facing cost overruns, and insulation upgrades were not in the budget. The modest goal of Energy Star Homes' basic score/rebates was at risk, but it seemed that little could be done. The project had nowhere to go for extra funds. The unit sales prices were set through the city of Boston's affordable artist live/work program formulas, so we couldn't increase the prices of the units. Many energy efficiency incentive programs are available in Massachusetts, and one of the services that my company, Urban Habitat Initiatives, provides is to help developers identify and combine programs to maximize benefits. This project proved to be a difficult fit. It was not clear how to find the resources the project needed to meet its energy efficiency goals. The project was complicated by the following:

No extra help from Energy Star Homes. The Massachusetts program provides additional grants to affordable housing projects, but it sets its affordability limit at 60% of the area median income for at least half of the units, with the remainder affordable to moderate-income households. Although Brookside has 18 units out of 24 set aside for affordable artist housing, the affordability ceiling for 15 of those units, much more than half, is 80% of median income.

No direct utility rebates or grants. KeySpan, the gas utility, provides rebates through the residential program for efficient furnaces or boilers for individual units, but the Rinnai units selected did

not qualify for the program, which has a minimum furnace efficiency requirement of 90% AFUE. As mentioned above, finding the best system was challenging, due to building constraints and right-sizing for small live/work units. The Rinnai systems were cost-effective to install and could be sized properly. However, their nominal efficiency did not meet the rebate program requirements, although the technical experts thought that the effective efficiency of the Rinnai units would probably exceed 90% AFUE furnaces, which would be oversized for the units and therefore wouldn't operate at full efficiency. KeySpan can also provide rebates and supplemental grants through its commercial programs, but because this project had unit-based rather than common heat and hot water, the commercial program did not apply.

A New Application of an Alternative Resource

Then the light bulb (CFL, of course) went on. After several conversations with KeySpan, we realized that since this project was existing housing, it could meet the requirements for KeySpan's residential weatherization program. Under the program, 20% of the installed cost of the approved measures could be rebated. The KeySpan program generally supports energy efficiency measures taken by individual homeowners, but the program staff saw the benefits of supporting a developer's efforts to achieve better energy performance for condominiums that would be sold to individual owners. They made the case internally and received approval from program managers to use funds this way.

Once approval to use the weatherization program in this innovative way was given, NADC was able to upgrade the insulation. The exterior wall insulation was changed to damp-spray cellulose, while the insulation for the underside of the floors (the ceiling of the dirt-floored crawlspace) was upgraded to Icynene. The damp spray cellulose provided better insulation, better sound control, and much better air sealing, and it eliminated the need for a vapor barrier. The Icynene between the crawlspace and the live/work spaces should keep the old industrial dust and



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higher REM:Rate scores on its new construction projects, but for Brookside it was a stretch to aim for basic qualification. Everyone agreed that passing was a victory—for energy efficiency, healthy housing, cost-effective live/work spaces, and for the new residents. The technical support from the Energy Star Homes program staff and program support from KeySpan through the innovative application of weatherization funds to the project enabled the development team to achieve its building performance goals.

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bacterial contaminants from being pulled up through the old floorboards and then breathed in by residents.

One challenge in using the KeySpan program was the requirement that the work be done by certified contractors. As it happened, the cellulose installer was a previously approved KeySpan contractor, but the Icynene installer was not. He had to become certified through the program before he could install the insulation. The new rigid insulation under the new roof also should have qualified for rebates under the weatherization program, but the installer was not certified and there was not enough time for him to become certified before the roof insulation was installed. As a result, the developer could not include this work in the rebate request. This presents a cautionary tale for widespread adoption of a single-family-oriented program to a multifamily development.

Multifamily general contractors usually price a job using a known set of subcontractors. When a certified list is mostly subcontractors that work in a single-family rather than a multifamily context, the general contractor may not be familiar with them and may not be comfortable including them on the construction team. Conversely, subcontractors who work primarily on multifamily projects may not

want to go through the certification process just to help the developer collect rebates, when they don't see opportunities for future business.

Costs

The energy efficiency upgrade costs were minimal, but they were critical to the feasibility of the project. The weatherization program rebates were just enough to cover the cost difference, and they made it possible to proceed. For phase 1, the cost of fiberglass insulation for the areas where cellulose and Icynene were installed was \$11,474. The upgrade cost was \$14,292, for a total increased cost of \$2,818. The weatherization program offered 20% of the total installed cost, or \$2,858, leaving the developer slightly ahead on a direct-cost basis. Also, because the insulation upgrades made the difference in the Energy Star program rating, the developer obtained Energy Star Homes program rebates of \$300 per unit.

End Result: We Made It!

Much to everyone's relief, the project passed the Energy Star Homes airtightness testing, and the project qualified for the basic program certification and rebate level. NADC typically achieves much

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